



山东高速路桥国际工程有限公司
SHANDONG HI-SPEED ROAD&BRIDGE INTERNATIONAL ENGINEERING CO.,LTD.



Expression of Interest

Ref: 2026-JYKF-UG-0427

Date: April 27th, 2026

To: Permanent Secretary
Ministry of Works and Transport
Kampala, Uganda

Re: EXPRESSION OF INTEREST IN THE REHABILITATION OF HOIMA-KAISO-
TONYA ROAD USING ASPHALT RECYCLING TECHNOLOGY

Dear Sir,

We write to respectfully express our strong interest in supporting the Ministry of Works and Transport in addressing the current condition of the Hoima-Kaiso-Tonya Road corridor.

We understand that sections of the road are experiencing deterioration, which may require substantial intervention. In this regard, our company proposes to deploy advanced asphalt recycling technology to rehabilitate the pavement through the reuse of the existing asphalt layers, blended with new asphalt materials, and subsequently re-laid to achieve improved structural performance.

This approach offers several key advantages:

- Significant reduction in overall rehabilitation costs compared to full-depth reconstruction;
- Efficient utilization of existing pavement materials, thereby minimizing material waste;
- Shortened construction timelines and reduced traffic disruption;
- Extension of the service life of the existing roadway through enhanced structural integrity.

We respectfully request the Ministry to provide the available technical documentation for the Hoima-Kaiso-Tonya Road, including but not limited to pavement design data, traffic studies, previous maintenance records, and any recent condition assessment reports. Upon receipt of

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1. CNR
He dle.
2. ADDNE
Review and write
them for
technical
meeting
13/05/26



EMU STEVEN
Please handle
15/05/26

Brenda/Rachael
Please handle
19/05/26



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the relevant information, our technical team will undertake a formal site inspection and engineering assessment to determine the most appropriate rehabilitation methodology, estimated project cost, and proposed implementation schedule.

We remain fully committed to working closely with the Ministry to develop a cost-effective, sustainable, and technically sound solution that aligns with Uganda's infrastructure development objectives.

Attached is a brief description of asphalt recovery technology and opinions on road repair made by our team on site in January.

We look forward to your favorable consideration and stand ready to engage at your earliest convenience.



Yours faithfully,

Alan Wu

Business Manager

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沥青路面再生技术

ASPHALT PAVEMENT RECYCLING TECHNOLOGY

公路交通节能与环保技术及装备
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Research and Development Center on
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Energy Conservation and Environment Protection
Ministry of Transport, PRC

中华人民共和国交通运输部

公司多年来致力于沥青路面再生方面的科技研发与技术创新,承担国家交通运输部公路交通节能与环保技术及装备交通运输行业研发中心核心任务,拥有省级创新型研究单位——山东省沥青路面再生工程技术研究中心,是全国唯一参编交通运输部《公路沥青路面再生技术规范》的施工企业。在就地热再生、就地冷再生、厂拌冷再生和厂拌热再生四项沥青路面再生技术及装备领域,公司自主创新、研发新设备,打破国外设备垄断,突破多项关键技术,形成自主核心技术与自主知识产权,实现制造产业化。多项沥青再生成果达到国际领先水平,获国家级和省部级工法 11 项;国家级和省部级奖励 18 项;实用新型及发明专利 36 项。目前已在山东、天津、内蒙古、江苏、浙江、新疆、湖南、云南、贵州、河南、福建等十几个省市相继应用,再生规模已累计达到 1500 万平方米。

The Group is committed to the research and development of science and technology of asphalt pavement recycling for years, and undertakes the core task of research and development center of energy saving and environmental protection technology for highway transportation and equipment transportation industry of Ministry of Transport of PRC. The Group also owns the provincial innovative research unit: research center of asphalt pavement regeneration engineering technology. Meanwhile, the Group is the only one of the construction enterprise participating in the preparation of Technical details for Asphalt Pavement Recycling of Ministry of Transport. As for the asphalt pavement recycling technologies of hot in-place recycling, cold hot in-place recycling, plant mixing cold recycling and plant heat recycling as well as related equipment field, the Group made breakthroughs to break the monopoly of the foreign markets, obtaining numbers of independent core technologies and intellectual property rights to realize the manufacturing industrialization. Lots of asphalt recycling achievements reached the international advanced level. The Group has won eleven construction processes of national and provincial and ministerial level and eighteen awards of national and provincial and ministerial level, and thirty-six utility model and invention patents. Right now the above-mentioned technologies have been applying in provinces of Shandong, Tianjin, Inner Mongolian, Jiangsu, Zhejiang, Xinjiang, Hunan, Yunnan, Guizhou, Henan, Fujian to reach fifteen million square meters of recycling scale.

沥青路面就地热再生技术

ASPHALT PAVEMENT HOT IN PLACE RECYCLING TECHNOLOGY

采用专用的热再生设备，对沥青路面进行加热、翻松，就地添加一定数量的新沥青、新沥青混合料、再生剂等，经热态拌合、摊铺、碾压等工序，一次性实现对路表一定深度范围内的旧沥青路面再生的技术。能够实现旧沥青混合料的 100% 利用，施工速度快、交通影响小，2-3 小时后即可通行。

自主研发的国内首套多步法沥青路面就地热再生机组，是国内唯一的多步法热风加热、分层铣刨再生机组，一举打破此类设备依赖于进口的行业垄断，整体创新技术达到“国际先进”水平，在多步热风加热、分层铣刨方面达到“国际领先”水平。

Heating and loosening the asphalt pavement, adding a certain amount of new asphalt, new asphalt mixtures and regenerant on the spot by means of hot mixing, paving, rolling and other processes via special hot recycling equipment, and old asphalt pavement recycling within the certain depth of the road can be realized at one time. Utilization of old asphalt mixtures can be completely achieved with fast construction speed and little traffic impact, and road will be passable in 2-3 hours.

The first self-developed domestic multi-step asphalt pavement hot in-place recycling unit is the only multi-step hot air heating and layered milling recycling unit in China. It breaks the monopoly of such equipment relying on import, and the overall innovation technology reaches "the international advanced level", and also has reached the "international leading level" in multi-step hot air heating and layered milling.



技术成熟的沥青路面就地热再生养护施工，实现高速公路养护零排放

The technically mature asphalt pavement hot in-place recycling maintenance construction can achieve zero emissions of highway maintenance.

沥青路面厂拌冷再生技术

ASPHALT PAVEMENT COLD MIX PLANT RECYCLING TECHNOLOGY

将沥青路面回收料 (RMAP) 运至拌和厂, 经破碎、筛分, 以一定比例与新集料、再生结合料、填料 (矿粉等)、水进行常温拌和, 常温铺筑形成路面结构层的沥青路面再生技术。

自主研发的厂拌冷再生设备, 具备精确的计量装置、供水系统, 独特的乳化沥青和泡沫沥青双供给系统以及二级强制搅拌等功能, 性能稳定, 生产能力大, 再生混合料质量更优。其中, 沥青发泡系统获国家实用新型专利, 填补国内空白。

Reclaimed materials from asphalt pavement (RMAP) are in a certain proportion mixed with the new aggregate, binder recycling and water at the normal temperature to form the pavement structure layer after transported to the asphalt mixing station for crushing and sieving.

The Group independently researched and developed the cold mix plant recycling equipment which has the precise metering device, water supply system, emulsified asphalt supply system, foamed asphalt supply system featuring the stable and productive performance, the high quality of recycled mixture. Especially, the asphalt foaming system won the national utility model patent which can fill the gaps in China.



自主研发的厂拌冷再生成套设备

Self-developed cold mix plant recycling complete sets of equipment



Suggestions on road repair in Hoima-Kaiso

(Shandong)

On January 23, 2026, the technical team went to the site of Hoima-Kaiso road to investigate the road conditions. There were longitudinal cracks on the asphalt surface of some sections, and the cracks were serious near Kaiso. The general situation is as follows:

I. Road sections with surface cracks

1、 There is a wildlife protection inspection station about 17 kilometers away from Kaiso. On the road between the station and Kaiso, the pavement structure is a two-layer asphalt pavement directly constructed on the gravel soil subgrade. The total thickness of the asphalt pavement is about 10cm, and longitudinal cracks appear regardless of the road sections, with uneven width. The edge of the subgrade is heavier than the center line of the subgrade. The asphalt concrete surface is finely graded, and some cracks are treated by asphalt potting, but the treatment is not thorough. The following figure

		
Longitudinal cracks in asphalt concrete pavement	The cracks at the edge are serious.	Thickness of asphalt concrete pavement
		
Asphalt concrete gradation is fine.	Asphalt joint filling treatment	crack width

2、 Analyze the causes

(1) The strength of the base is insufficient, resulting in uneven settlement. The compaction degree of gravel soil is insufficient, and the gradation of granular materials is uneven. Under the repeated action of traffic load, the layers of the pavement locally sink and deform unevenly, and the asphalt concrete surface layer is "pulled" to form longitudinal cracks.

(2) During the construction process, due to the reasons of gradation, non-layered rolling or

		
Pavement cracking	The original ground slope is relatively large.	Pavement crack
		
Pavement crack	Pavement crack thickness	Pavement structure thickness

2、Cause analysis

(1) The original ground has a large transverse slope, and if it is not layered in the construction process, the subgrade will easily slip, and the adhesion between asphalt pavement and graded crushed stone is poor, which will lead to the slip of pavement surface under the action of vehicle load and form longitudinal cracks.

(2) The thickness of asphalt concrete pavement is about 15cm, and it is directly paved on graded crushed stone, so the pavement strength is seriously mismatched with the bearing capacity of the base.

(3) There are fine mesh cracks at the edge of cracks and weeds grow in cracks, which shows that the long-term infiltration of rain has led to the softening of base and subgrade, forming a vicious circle of "cracking → water seepage → more cracking"

(4) Construction defects, the thickness of asphalt surface layer is about 15cm, which is not paved in layers on site, and the rolling thickness is too large to be effectively compacted.

(5) The shoulder is seriously damaged, and the water seepage at the edge is serious, which further leads to subgrade damage and pavement cracks.

3、Processing method

(1) Milling and planing the diseased section: planing the asphalt surface in the cracked section,

and cleaning the loose base and subgrade.

(2) Subgrade treatment and strong replacement and filling of base course: the soft gravel soil base course and subgrade shall be excavated, and the subgrade shall be filled by benching and filling and compacted in layers, with the compactness not less than 96%. Cement stabilized crushed stone can be used as the base course material in this section.

(3) Interlayer treatment: spray prime coat on the top of the base, and after drying, geogrids can be added to improve the crack resistance.

(4) Asphalt pavement: asphalt pavement shall be paved in two layers, and the degree of compaction shall not be less than 98%.

(5) Optimize drainage and shoulder protection: cement concrete slope protection shall be made on the left side to prevent surface water from invading the subgrade on the hillside, and back pressure berm shall be made on the right side to prevent the subgrade from slipping.

4、Key measures to prevent recurrence

(1) Strictly implement the bonding process between layers, spray prime coat on the base, spray tack coat between the surface layers, and lay geosynthetics when necessary.

(2) Strengthen the inspection of the construction process. When the cross slope is more than 5%, construct it by benching and filling, and the compactness of the base and the surface must be strictly controlled.

(3) The construction thickness of each layer of asphalt should not be too large, so as to avoid insufficient stress concentration compaction caused by too large a single layer thickness.